

JOHNS HOPKINS
WHITING SCHOOL of ENGINEERING

AI Ethics and Social Impact

601.770

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How can we develop AI for good and not for bad?

Decisions we make about our data, methods, and tools are tied up with their impact on people and societies



People create data



People use models

Data

Data Labels

Models

Model Outputs

Output Usage



People build models



People are affected by models

Example: Are there some applications we should not build?

Hypothetical case: should we build a classifier to predict someone's sexual orientation from their photo?

Sexual Orientation Classifier

Who can be harmed by such a classifier?

- Personal attributes (gender, race, sexual orientation, religion) are complex social constructs, not categorial/binary, are dynamic, are private and often not visible publicly
- These are properties for which people are often discriminated against
  - In many places being gay is prosecutable
  - Such a classifier might affect people's employment, family relationships, health care opportunities, etc.

Additional Ethical Questions

- Who can benefit from such a classifier?
- Where does the training data come from?
- Did anyone consent?

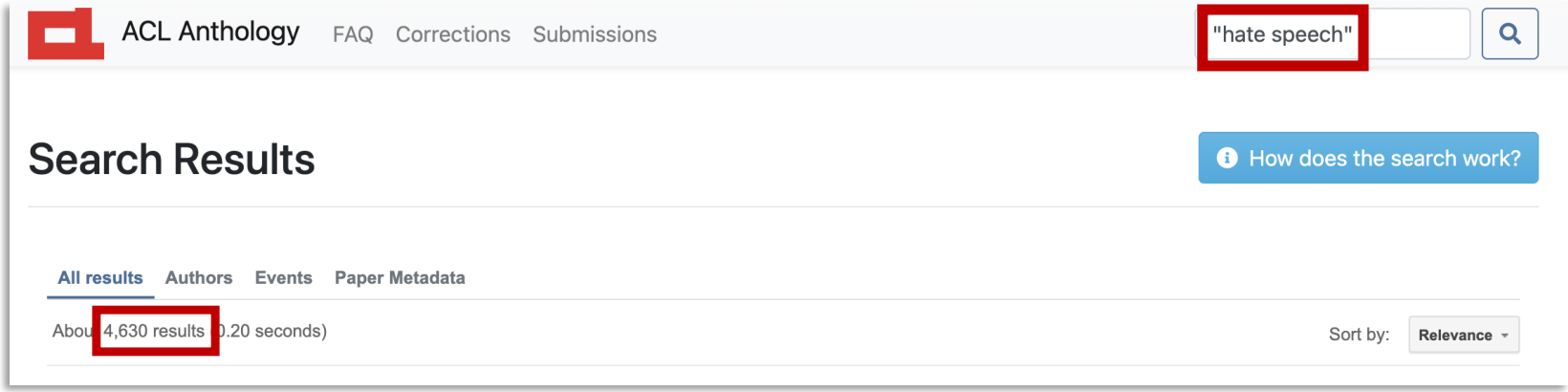
Most examples are not so straightforward

Problem:

- Hate speech and offensive language are prevalent on the internet and can lead to tangible harms
- Marginalized people are disproportionally targets of hate speech
- Manually identifying hate speech is difficult for human moderators
  - Too much volume to keep up with
  - Mental toll of reading offensive content

Technical Solution

- Build NLP models to identify hate speech automatically

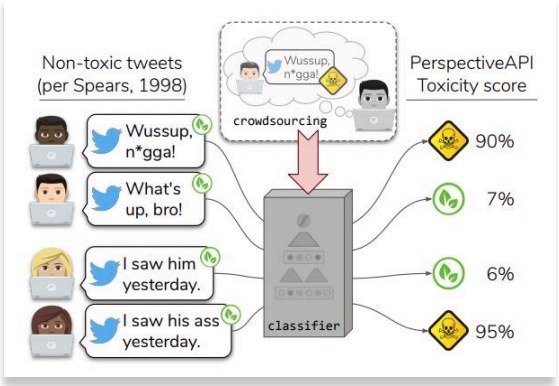


More Problems: hate speech detectors are biased

Term	Toxic	Overall
atheist	0.09%	0.10%
queer	0.30%	0.06%
gay	3%	0.50%
transgender	0.04%	0.02%
lesbian	0.10%	0.04%
homosexual	0.80%	0.20%
feminist	0.05%	0.05%
black	0.70%	0.60%
white	0.90%	0.70%
heterosexual	0.02%	0.03%
islam	0.10%	0.08%
muslim	0.20%	0.10%
bisexual	0.01%	0.03%

Table 1: Frequency of identity terms in toxic comments and overall.

[Dixon et al. 2018]



[Sap et al. 2019]

Even more problems

- How do define what is offensive or hate speech?
  - Norms differ widely in different communities
  - Setting a universal standard is enforcing a majority viewpoint
- Who has control of the technology?
  - Concentrating power in few hands
- How might this technology be abused? (dual use potential)
  - Hate speech generator
  - Censorship

Solutions?

- Don’t build AI/NLP for hate speech detection?
- But then what about all the hate speech on the internet?
- Maybe we should ban social media? The internet?

This course will often be about *asking questions* and not always *finding answers*

This Course

- Case studies of ethical challenges in AI research and development
  - Part 1 structured more around ethical challenges (data and privacy, fairness, etc.)
  - Part 2 structured more around applications (social services, criminal justice, healthcare)
  - Some readings are more technical, many are not

- Frameworks and guidelines
  - Code of ethics, policy and regulation

- Identifying and discussing series of questions
  - ~~What is AI?~~
    What should AI be? [Virginia Dignum, Interspeech Keynote 2023]

Course website

- https://ai-ethics-601-770.cs.jhu.edu/fa2026/

- Current plan: Announcements through Canvas

Course Format

- Before class:
  - Read 3-4 papers on 2 topics (~2 papers per topic)
  - Discussion leaders:
    - Decide on questions and topics for discussion during the class period
    - Send discussion plan to Anjalie and Krithika by 5pm Friday
- During class (**in person**):
  - For each topic:
    - 10min: Individual journal reflection on the assigned readings
      - Don't be late!
    - 45min small discussion groups (6-7 students)
      - Discussion leaders
    - 15min: Discussion leaders share back to the entire class
- Occasionally we will deviate with a lecture or guest

Course Project

- Recommended groups of 3ish people
- Can be on any topic related to the course:
  - Technical paper
  - Survey/position paper

Grading

- Coursework (40%):
  - Reading responses (15%)
  - Class participation (25%)
- Project (55%):
  - Project literature review (15%)
  - Project written proposal and presentation (15%)
  - Project final presentation and paper (20%)
  - Final Project Q&A (10%)

Learning Objectives

1. Develop paper reading skills

2. Develop ability to think critically about ethical issues in AI development and use
   a) Leading and participating in discussions on these topics
   b) Identifying and characterizing problems
   c) Brainstorming ways to move forward

3. Build domain knowledge about ethical issues studied in AI
   a) Focus on reading primary sources (research papers, policy documents, etc.) over media-hype stories

Today's topics

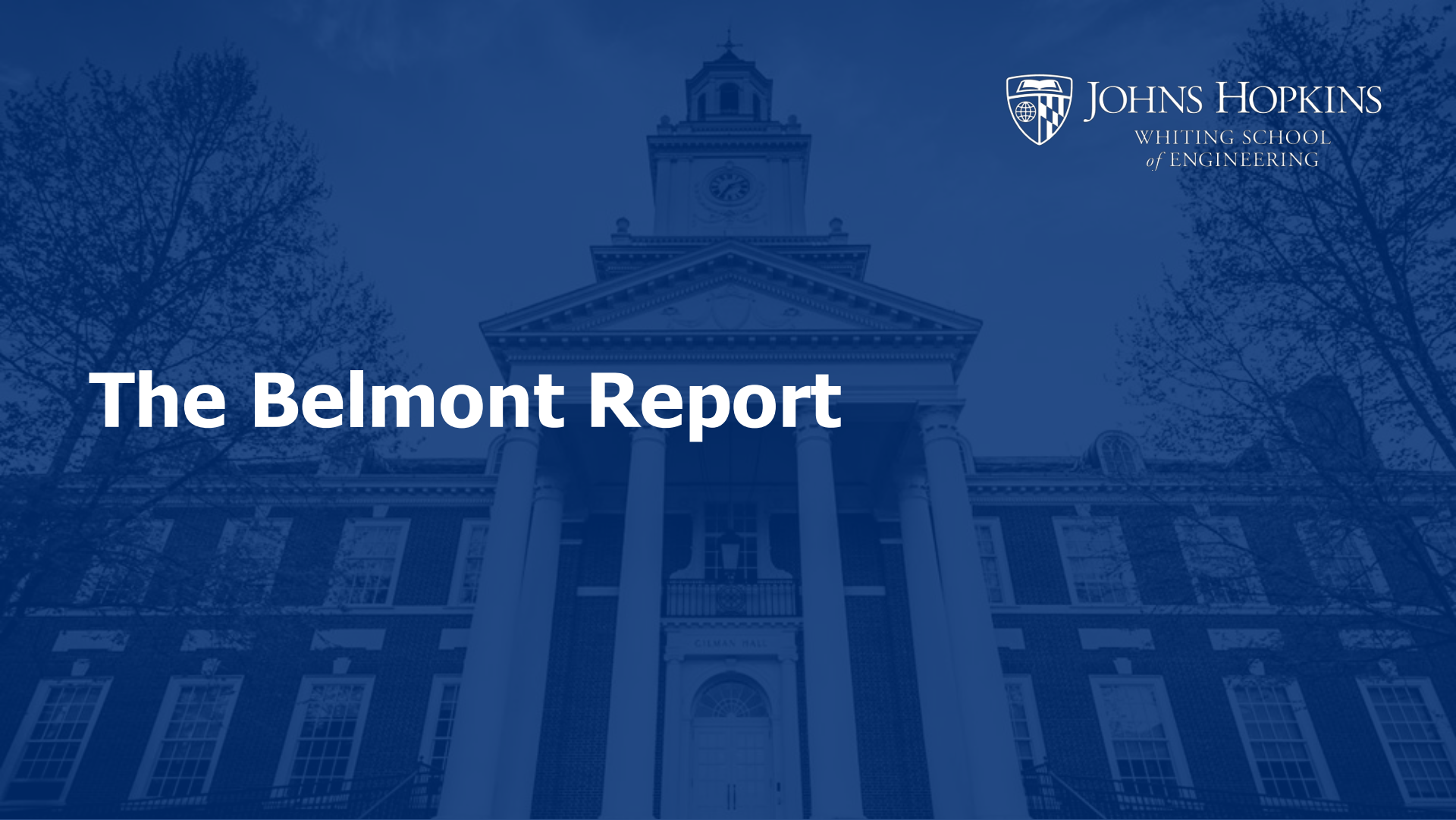
Introduction:

- Origins of research ethics
- Brief history of human subjects protection

Model development pipelines: Data ownership

- Brief background and reading break

Practice discussion



The Belmont Report

Nuremberg Code of 1947

- Ten principles of research developed for the "Doctors' Trial": American judges trying Nazi doctors accused of murder and torture in their human experiments in the concentration camps.

- Highlights:
  - 1. The voluntary consent of the human subject is … essential.
  - 2. The experiment should be …. for the good of society
  - 6. …risk ... should never exceed ... the humanitarian importance of the problem
  - 9. ...subject should be at liberty to bring the experiment to an end...

Shuster, Evelyne. 1997. "Fifty years later: the significance of the Nuremberg Code." *New England Journal of Medicine* 337, 20: 1436-1440.

United States Public Health Services Study in Tuskegee

- 40-year study by the US Public Health Service begun in 1932
- Goal: observe natural history of untreated syphilis
- Enrolled 600 poor African American sharecropper men
  - 400 with syphilis, 200 controls
- Told they would be treated for "bad blood"
- Were not treated, merely studied
  - Were not told they had syphilis
  - Sexual partners not informed
  - By 1940s penicillin becomes standard treatment for syphilis
    - Subjects were not told or given penicillin

Wikimedia Commons, from National Archives



United States Public Health Services Study in Tuskegee

- 1964 Protest letter from a doctor who reads one of the papers
  - "I am utterly astounded by the fact that physicians allow patients with a potentially fatal disease to remain untreated when effective therapy is available"

- 1965 Memo from authors:
  - "This is the first letter of this type we have received. I do not plan to answer this letter"

United States Public Health Services Study in Tuskegee

- 1966 Peter Buxtun, a PHS researcher in San Francisco, sent a letter to the CDC but study was not stopped.
- 1972 Buxtun goes to the press.
- Senator Edward Kennedy calls congressional hearings
- 1974 Congress passes National Research Act



Syphilis Victims in U.S. Study Went Untreated for 40 Years

By JEAN HELLER
The Associated Press

WASHINGTON, July 25—For 40 years the United States Public Health Service has conducted a study in which human beings with syphilis, who were induced to serve as guinea have serious doubts about the morality of the study, also say that it is too late to treat the syphilis in any surviving participants.

Doctors in the service say

NY Times July 26, 1972

National Research Act 1974

- Required institutional review of all federally funded experiments
  - Institutional Review Boards (IRBs)
- Created National Commission for the Protection of Human Subjects of Biomedical and Behavioral Research
  - Issued Belmont Report in 1976/1979
- The Common Rule: Title 45, Part 46 of the Code of Federal Regulations: Protection of Human Subjects.
  - Informed consent

What falls under IRB jurisdiction?

- Operate under federal (U.S. government) requirements
  - Don’t cover non-U.S. institutions
- Institutions receiving federal support
  - Companies that don’t receive federal grants may have internal review processes but are not mandated to have IRBs
- “Human subject research”
  - Biospecimens
  - Identifiable private information

https://homewoodirb.jhu.edu/
https://www.hopkinsmedicine.org/institutional_review_board/

The Belmont Report

"to identify the basic ethical principles that should underlie the conduct of biomedical and behavioral research involving human subjects and to develop guidelines which should be followed to assure that such research is conducted in accordance with those principles"

A. Boundaries Between Practice and Research
B. Basic Ethical Principles
C. Applications

An ethical framework that offers very practical and applied guidance

https://www.hhs.gov/ohrp/regulations-and-policy/belmont-report/read-the-belmont-report/index.html

The Belmont Report: Respect for Persons

1. "Individuals should be treated as autonomous agents."
2. "Persons with diminished autonomy are entitled to protection"

Example of 1:

- People should engage in research voluntarily and with informed consent

Example of 2:

- IRBs required increased scrutiny for studies involving vulnerable populations like people in prison

The Belmont Report: Beneficence

"Do no harm. Maximize benefit and minimize risk"

- Example of failure: Tuskegee syphilis experiment

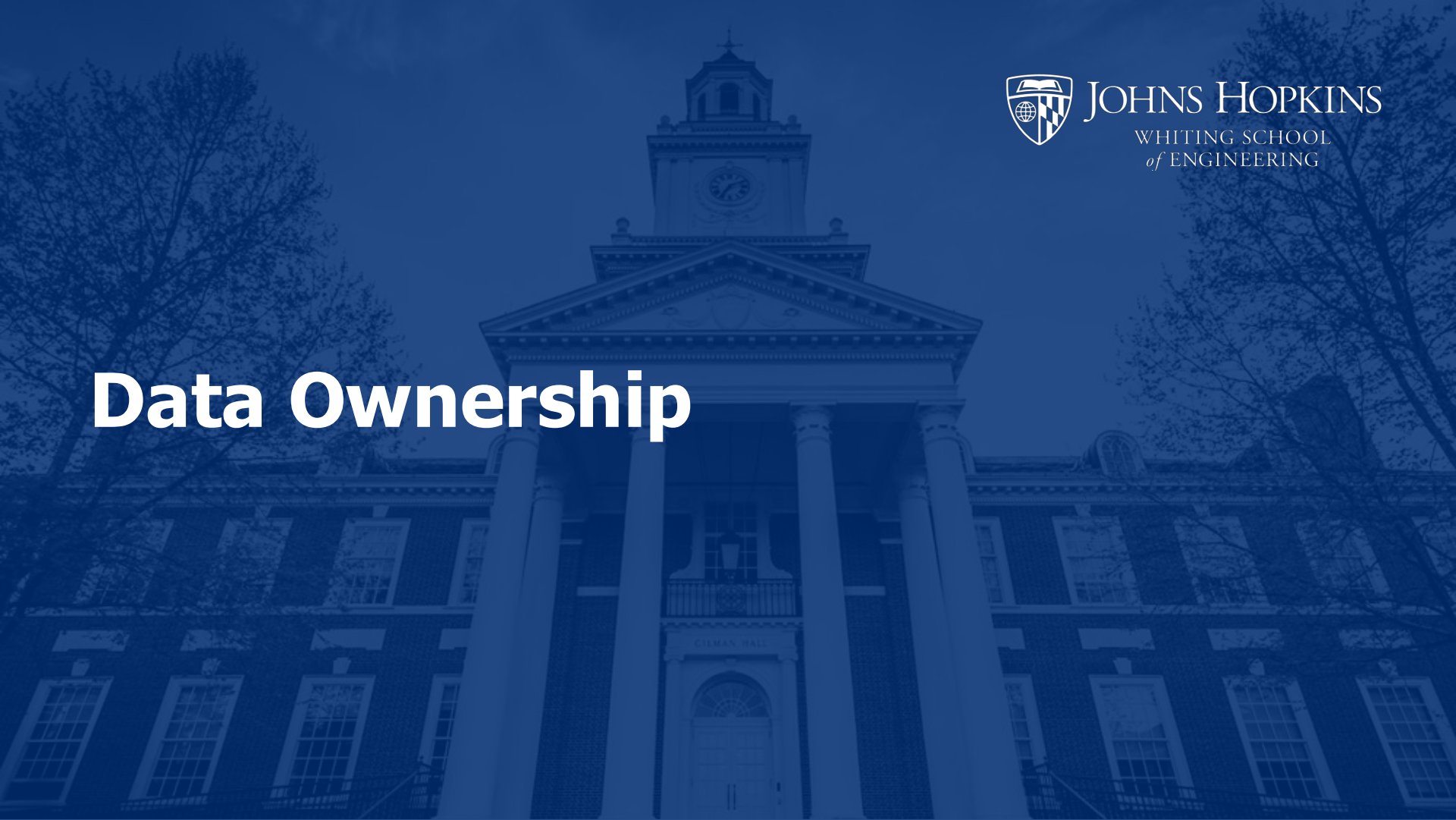
The Belmont Report: Justice

- “fairness in distribution" or "what is deserved.”
- “The benefits and risks shared by a population that may benefit from the results of research”
- “Another way of conceiving the principle of justice is that equals ought to be treated equally”

Example failure: unwilling prisoners as research subjects in Nazi concentration camps

The Belmont Report: Applications

- Informed consent
  - Information, Comprehension, Voluntariness
- Assessment of Risks and Benefits
- Selection of Subjects



Data Ownership



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Brief History of Facial Recognition Research

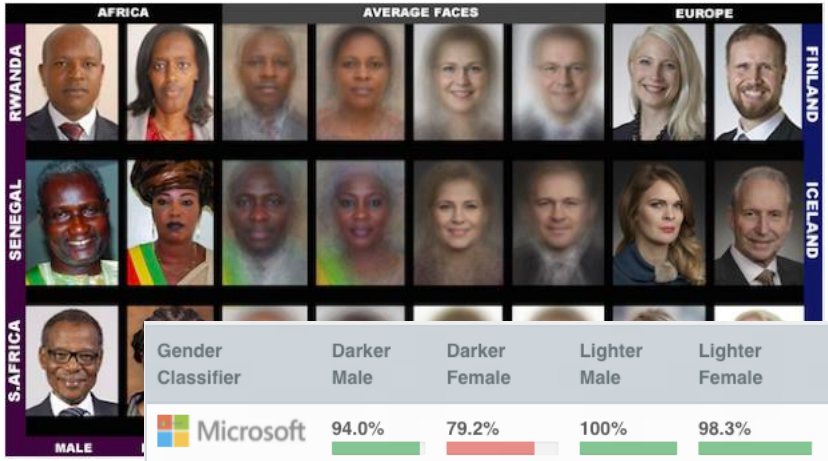
- 1960s: Woody Bledsoe, Helen Chan Wolf, and Charles Bisson try to get computers to recognize faces
- 1993-1996: DARPA FERET project
  - "to develop automatic face recognition capabilities that could be employed to assist security, intelligence, and law enforcement personnel in the performance of their duties"
- 1996-2006: FERET dataset simulates research interest, but models are not very good
- 2007-2013: Mainstream development for unconstrained settings
  - Hunt for "in the wild data sets"
- 2014-present: Deep learning leads to more accurate models and commercialization

https://www.nist.gov/programs-projects/face-recognition-technology-feret

Raji, Inioluwa Deborah, and Genevieve Fried. "About face: A survey of facial recognition evaluation." AAAI 2020 Workshop on AI Evaluation (2021).

Gender shades: Intersectional accuracy disparities in commercial gender classification

- Popular benchmark datasets are overwhelmingly composed of lighter-skinned subjects
- Evaluation of 3 commercial gender classification systems shows that darker-skinned women are by far the most misclassified group



Gender Classifier	Darker Male	Darker Female	Lighter Male	Lighter Female	Largest Gap
Microsoft	94.0%	79.2%	100%	98.3%	20.8%
FACE++	99.3%	65.5%	99.2%	94.0%	33.8%
IBM	88.0%	65.3%	99.7%	92.9%	34.4%

Buolamwini, Joy, and Timnit Gebru. "Gender shades: Intersectional accuracy disparities in commercial gender classification." FAccT. PMLR, 2018.

“Diversity in faces” data set

- Proposed Solution:
  - We need to train and evaluate models on more diverse data sets

- “Diversity in faces” data set, containing more diverse images annotated with features
  - Data was collecting from an existing ML data set of faces, originally collecting from Flickr, a photo sharing website
  - Annotated data for various features including facial geometry and skin-tone



https://www.ibm.com/blogs/research/2019/01/diversity-in-faces/

“Diversity in faces” data set

- “This is the dirty little secret of AI training sets. Researchers often just grab whatever images are available in the wild,” - NYU School of Law professor Jason Schultz
- “Despite IBM’s assurances that Flickr users can opt out of the database, NBC News discovered that it’s almost impossible to get photos removed. IBM requires photographers to email links to photos they want removed, but the company has not publicly shared the list of Flickr users and photos included in the dataset”
- Lawsuits over this use of data filed by people whose photos were in it
- IBM took down the data set



https://www.nbcnews.com/tech/internet/facial-recognition-s-dirty-little-secret-millions-online-photos-scraped-n981921

Impacts: Where is facial recognition technology not used?

2018:

- Gender Shades Project
- Intense criticism of facial recognition research as surveillance tech (petitions, letter to Jeff Bezos)

2020:

- Amazon places 1-year moratorium on use of it's face-processing software by police agencies (which they extended at least one more year)
- Microsoft stops selling facial-recognition software to police
- IBM halts work on face recognition because it's used for racial profiling

https://www.technologyreview.com/2020/06/12/1003482/amazon-stopped-selling-police-face-recognition-fight/
https://www.washingtonpost.com/technology/2020/06/11/microsoft-facial-recognition/
https://www.technologyreview.com/2020/06/09/1002947/ibm-says-it-is-no-longer-working-on-face-recognition-because-its-used-for-racial-profiling/

How is facial recognition technology used?

Ad from a face image processing company

> "We live in a dangerous world, where harm doers and criminals easily mingle with the general population; the vast majority of them are unknown to the authorities….Public Safety agencies, city police department, smart city service providers and other law enforcement entities are increasingly strive for Predictive Screening solutions, that can monitor, prevent, and forecast criminal events and public disorder without direct investigation or innocent people interrogations.

> "What if it was possible to know whether an individual is a potential pedophile, an aggressive person, or a criminal?"

How is facial recognition technology used?

- Emotion expression and intensity
- Sexual orientation
- Political orientation
- Personality traits

A large body of work in psychology and anthropology has provided evidence that "there is no relationship between how we look and how trustworthy or intelligent we actually are", though opposing research exists

Severin Engelmann, Chiara Ullstein, Orestis Papakyriakopoulos, and Jens Grossklags. 2022. What People Think AI Should Infer From Faces. (FAccT '22) https://doi.org/10.1145/3531146.3533080

Journal reflection and Break

- Briefly introduce yourself
- Why are you taking this class?
- What do you hope to learn?

- Discussion groups:
  - We will start with asking everyone to introduce themselves and share a few thoughts on the topics presented today

Class recap

- **Not** intended to be a summary of the entire discussion, instead, it should be the **outcomes** of your discussion



Next class

Data Privacy

Data Annotation

Data Privacy

1. Lundberg, Ian, et al. "Privacy, ethics, and data access: A case study of the Fragile Families Challenge." Socius 5 (2019)
   a) A case studying designing steps that one research group took to mitigate data privacy risks in a shared task
   b) Nice overview of security-focused thinking (threat models, adversaries)
   c) Example of how one could be careful about data privacy
2. Mireshghallah, Niloofar and Tianshi Li "Position: Privacy Is Not Just Memorization!", 2025
   a) Privacy in risks in LLMs
   b) Pay particular attention to Sections 2 and 3

Note: both readings are a little longer than usual, but they are not very technical

Data Annotation

- The development of Generative AI models (LLMs) involves extensive use of *annotated data:*
  - Pre-training data needs to be filtered for quality and toxicity
    - One of the critical use cases of content moderation annotations is to build classifiers **to identify and remove toxic content from training data**
  - Models are instruction-tuned and trained with reinforcement learning using annotated data

1. Michelle Du and Chinasa T. Okolo, "Reimagining the future of data and AI labor in the Global South" Brooking Institute, 2025
   a) **Note**: one of the critical use cases of content moderation is filtering AI training data
2. Data Workers Inquiry
   a) https://data-workers.org/

Data Workers

1. Michelle Du and Chinasa T. Okolo, "Reimagining the future of data and AI labor in the Global South" Brooking Institute, 2025

2. Data Workers Inquiry
   a) https://data-workers.org/
   b) Browse some of the products

Data Workers Inquiry

- A research initiative that seeks to expose experience of data workers
- Co-production with data workers:
  - Data workers are provided resources and support for conducting research and reporting findings about their workplaces and the experiences of data workers
  - Products include zines, documentaries, comics, essays, podcasts, and animations

- Browse some of the products
  - Content warning: Many of the content involves subjects that may be disturbing, including graphic descriptions of annotated content, struggles with mental and addiction, etc.
  - STOP reading/watching if the content is harmful for you (not assigning a specific study for this reason)

Miceli, Milagros, et al. "Methodological Considerations for Centering Workers' Epistemic Authority in AI Research." *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society*. Vol. 8. No. 2. 2025.

Discussion Guide Expectations

- Standard starting point: for the first 10min, go around the group and encourage everyone to share a few thoughts about the readings
- Include questions and/or activities for discussion groups to conduct in class
  - Think about what might be valuable outcomes (learning objectives) from the discussion
  - What are aspects of the readings that are unclear / hard to apply that discussion might help clarify?
  - What do you want other students in the class to learn and understand from the readings?
- Include approximate time for each topic or activity: you're aiming for ~45min of discussion time
- You're encouraged to explore beyond the readings
  - What other information did you find that you might share with your group?

Discussion Groups

- Make sure that each person in your group is talking. If someone is talking less, find a way to draw them in. This is one of the most important parts of your role as facilitator: do your best to have everyone talk (very roughly) the same amount.
- Try not to respond to every comment that everyone makes at your table; instead if no one is talking ask the table what they think about what has just been said. If it feels like the right time to summarize responses, maybe do so without taking a stand one way or another.
- If you need an icebreaker, you might ask a question for which there is no single correct answer and go around the table with it. (Example: "On a scale of 1 to 10, how successful do you think the paper is at solving the problem they pose" or "If you were going to do a project following up on this paper, which part of the paper might be most rewarding to look into?")
- Try to be careful about giving credit for ideas

Closing logistics

- No class 9/7: Labor Day
- Next class (9/14):
  - Data Privacy
  - Data workers

- Decide on a discussion leader for next class